

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Turner et al.	2016	Nationwide	30-99		Annual	0.005827	0.000963	Log-linear	Single-pollutant model
Emergency Hospital Admissions, All Respiratory	Zanobetti et al.	2009	26 U.S. communities	65-99		D24HourMean	0.002049	0.000437	Log-linear	

E.7.1 Di et al. (2017)

See full study description under Di et al. (2017) in Appendix E, Section E.1.1.

Mortality, All-Cause (Single-Pollutant Model)

In a single-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.084) and 95% confidence interval of (1.081-1.086) associated with a change in annual mean PM_{2.5} exposure of 10.0 µg/m³ (Di et al., 2017, Table 2 Single-pollutant analysis).

Mortality, All-Cause (Nearest Monitor Analysis)

In a two-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.061) and 95% confidence interval of (1.059-1.063) associated with a change in annual mean PM_{2.5} exposure of 10.0 µg/m³ (Di et al., 2017, Table 2 Nearest Monitor Analysis, Cox PH with GEE).

Mortality, All-Cause (Cox Proportional Hazards Model with Mixed Effects (COXME))

In a two-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.081) and 95% confidence interval of (1.078-1.083) associated with a change in annual mean PM_{2.5} exposure of 10.0 µg/m³ (Di et al., 2017, Table S3 Main Analysis, Cox PH with mixed effects (COXME)).

E.7.2 Jones et al. (2015)

Jones et al. (2015) assessed the impacts of PM_{2.5} and its chemical constituents (sulfate (SO₄), ammonium (NH₄), nitrate (NO₃), elemental carbon (EC), and carbon-only portion of organic carbon aerosol) on respiratory health. The study encompassed all ages, races, and ethnicities with a case-crossover analysis in New York state. Analysis used 24-hour average PM_{2.5} chemical constituent concentrations from the Community Multiscale Air Quality (CMAQ) model, and meteorological data from the National Climatic Data Center. The authors assessed hospital discharge data from the New York State Department of Health State Planning and Research Cooperative System (SPARCS) through principle diagnosis categorized by ICD-9 code (chronic bronchitis (ICD-9 491), emphysema (ICD-9 492), asthma (ICD-9 493), and chronic airway obstruction (ICD-9 496)). Authors used a single pollutant conditional logistic regression model to analyze

the respiratory hospital admission and PM_{2.5} chemical constituent data over time and by season. The authors calculated hazard ratios (HRs) using the PHREG procedure in SAS (version 9.2) with 95% confidence intervals from the regression models. Jones et al. (2015) found that PM_{2.5} and its chemical constituents showed significant associations between total PM_{2.5} mass and hospital admissions in the year-round model and for all exposure lags (0-4 days). Of all the PM_{2.5} chemical constituents, sulfate had the strongest association with respiratory hospital admissions, particularly during the summer months. Additionally, sulfate was the largest contributor to the PM_{2.5} total mass (49.9%).

Hospital Admissions, Respiratory-1 (ICD-9 Codes 491, 492, 493, 496)

In a year-round single-pollutant model, the coefficient and standard error were estimated from a hazard ratio of 1.006 (95% CI: 1.003-1.008) for a 7.48 µg/m³ increase in daily mean PM_{2.5} concentrations, lagged by 4 days. The model was adjusted for season. (Jones et al. 2015, Figure 2).

E.7.3 McConnell et al. (2010)

McConnell et al. (2010) examined the association between long-term traffic-related air pollution (PM_{2.5}, PM₁₀, O₃, and NO₂) exposure and incident asthma in children. The authors collected data for three years from a cohort of 2,497 kindergarten and first-grade children who entered the Southern California Children's Health Study without asthma or wheeze. McConnell et al. (2010) defined new-onset asthma as physician-diagnosed asthma reported by parents on a yearly questionnaire. While the primary focus of the study was traffic-related air pollution from local vehicle emissions, the authors also utilized ambient air pollution exposure data from central site monitors in each of the 13 communities in the Southern California Children's Health Study. The authors used a multilevel Cox proportional hazards model to estimate the association between ambient air pollution exposure and new-onset asthma, controlling for race/ethnicity, secondhand smoke exposure, and pets in the home. The authors concluded that traffic-related pollution exposure may contribute to an increased risk of new-onset asthma in children.

Incidence, Asthma

In a single-pollutant model, the coefficient and standard error were estimated from a hazard ratio of 1.66 (95% CI: 0.91-3.05) for a 17.4 µg/m³ (range of exposure in the 13 communities) increase in annual average PM_{2.5} exposure (McConnell et al. 2010, Table 4).

E.7.4 Nishimura et al. (2013)

Nishimura et al. (2013) investigated the relationship between long-term early-life pollution exposure (PM_{2.5}, PM₁₀, O₃, NO₂, and SO₂) and asthma onset in Latino and African American children in five urban areas (Chicago, IL; Bronx, NY; Houston, TX; San Francisco, CA; Puerto Rico). The authors obtained data from the Genes-environments and Admixture in Latino Americans (GALA II) Study and the Study of African

Americans, Asthma, Genes and Environments (SAGE II). GALA II and SAGE II are case-control studies that enrolled children with and without asthma. The studies defined case subjects as children with physician-diagnosed asthma plus two or more symptoms of coughing, wheezing, or shortness of breath in the two years before study enrollment while control subjects were children with no reported history of asthma, lung disease, or chronic illness, and no reported symptoms of coughing, wheezing, or shortness of breath in the two years before study enrollment. The authors estimated annual average pollution exposures during the first year of life as well as the first three years of life from self-reported residential histories by calculating inverse distance-squared weighted averages from the four closest U.S. EPA Air Quality System monitoring stations within 50 km. The authors first used regional- and study-specific logistic regression models to estimate the association between asthma diagnosis and pollution exposure, controlling for population subgroup status and subsequently combined the regression coefficients into a multi-region estimate using a random-effects meta-analysis. Nishimura et al. (2013) showed that early-life air pollution exposure may increase the risk for asthma development in later childhood for Latino and African American cohorts.

Incidence, Asthma

In a single-pollutant model estimating PM_{2.5} exposure during the first year of life, the coefficient and standard error were estimated from an odds ratio of 1.03 (95% CI: 0.90-1.18) for a 1 µg/m³ increase in average annual PM_{2.5} levels at the residential address during the first year of life (Nishimura et al. 2013, Figure 2).

E.7.5 Pope et al. (2015)

Pope et al. (2015) evaluated the relationship between long-term exposure to ambient PM_{2.5} and risk of death from CVD and cardiometabolic disease, including effect modification of the relationship by pre-existing cardiometabolic risk factors, in the ACS Cancer Prevention Study II cohort (ages 30+). PM_{2.5} exposures were estimated at home addresses based on a land use regression model with Bayesian Maximum Entropy kriging of residuals (LURBME). Pope et al. utilized a Cox proportional hazards model controlling for individual-level covariates which included variables that characterized current and former smoking habits, exposure to second-hand cigarette smoke, workplace PM_{2.5} exposure in each subject's main lifetime occupation, self-reported exposure to dust and fumes in the workplace, marital status, level of education, body mass index, consumption of alcohol, and quartile ranges of dietary fat index and quartile ranges of a dietary vegetable/fruit/fiber index. Ecological covariates included median household income; percentage of people with <125% of poverty level income; percentage of unemployed individuals aged ≥16 years; percentage of adults with <12th grade education; and percentage of the population who were Black or Hispanic.

Mortality, All-Cause (LURBME)

In a single-pollutant model, the coefficient and standard error are estimated from the hazard ratio (1.07) and 95% confidence intervals (95% CI: 1.06-1.09) for a 10 µg/m³

increase in monthly PM_{2.5} exposure levels averaged from 1999-2004 (Pope, et al., 2015, Table 1. Cox model with individual-level plus ecological covariates; exposure based on LUR-BME).

E.7.6 Talbott et al. (2014)

Talbott et al. (2014) assessed daily PM_{2.5} concentrations and hospitalizations for cardiovascular disease in Florida, Massachusetts, New Hampshire, New Jersey, New Mexico, New York, and Washington from 2001 to 2008. The authors gathered hospital discharge data from each state's respective data stewards. Talbott et al. (2014) conducted a time-stratified case-crossover study using hospitalization data for all cardiovascular disease (ICD-9 390-459) and for several specific cardiovascular diseases within the ICD-9 390-459 range. Authors used a downscaling Bayesian space-time modeling approach to combine air monitoring data and air gridded numerical outputs from the Community Multi-Scale Air Quality Model (CMAQ) to predict daily PM_{2.5} concentrations. The authors gathered meteorological data from the CDC Wonder North America Land Data Assimilation System Daily Air Temperatures and Heat Index. Talbott et al. (2014) used conditional logistic regression adjusted for O₃ (same day as PM_{2.5}) and maximum apparent temperature (same day as admission).

Hospital Admissions, All Cardiac Outcomes (ICD-9 Codes 390-459)

In a two-pollutant multivariable model with O₃, the coefficient and standard error are estimated from an odds ratio of 1.005 (95% CI: 0.998-1.012) for Massachusetts; 1.011 (95% CI: 1.007-1.016) for New Jersey; 1.011 (95% CI: 0.973-1.050) for New Mexico; 1.011 (95% CI: 1.008-1.014) for New York; 0.996 (95% CI: 0.990-1.002) for Florida; 0.988 (95% CI: 0.965-1.013) for New Hampshire; and 0.991 (95% CI: 0.981-1.002) for Washington. Each odds ratio is for a 10 µg/m³ increase in the averaged daily mean PM_{2.5} concentration 0-, 1-, and 2-day lags (Talbott et al. 2014, Table 3).

E.7.7 Turner et al. (2016)

See full study description under Turner et al. (2016) in Appendix E, Section E.1.2.

Mortality, All-Cause (Single-Pollutant Model)

In a single-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.06) and 95% confidence interval of (1.04–1.08) associated with a change of 10.0 µg/m³ in the mean PM_{2.5} exposure level from 1999-2004 (Turner et al., 2016, Table E10 HBM PM_{2.5}, 1982-2004).

E.7.8 Zanobetti et al. (2009)

Zanobetti et al. (2009) examined the relationship between daily PM_{2.5} levels and emergency hospital admissions for cardiovascular causes, myocardial infarction, congestive heart failure, respiratory disease, and diabetes among 26 U.S. communities from 2000-2003. The authors used meta-regression to examine how this association was modified by season- and community-specific PM_{2.5} composition while controlling